

2024 SCHEME

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MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT347				
Course Name: ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT				
Max. Marks: 50			Duration: 2 hours 30 minutes	
PART A				
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What is civic virtue? How does it relate to professional ethics?	1	(3)
2		What do you mean by 'gendered technology' and provide one specific example of a product whose design or application reflects this concept.	2	(3)
3		Explain the term triple bottom line in sustainable engineering.	3	(3)
4		Why is biodiversity considered essential for ecosystem stability? Give one example.	3	(3)
5		What is the idea behind degrowth? How is it different from economic development?	4	(3)
6		Identify two engineering solutions that can reduce greenhouse gas emissions in everyday life.	5	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9		You are part of a team designing an app that collects user data for public services. What ethical concerns would you raise regarding data privacy and consent? Suggest safeguards.	1	(8)
10	a)	What is the fundamental difference between 'Equity' and 'Equality' when discussing gender initiatives and resource allocation in education/workplace?	2	(5)
	b)	Suggest measures to promote gender equity in workplaces.	2	(3)
Module 2				
11		A new industrial project near a wetland promises jobs but threatens local ecosystems. Apply the three ethical lenses (anthropocentric, biocentric, ecocentric) to assess it and recommend a balanced decision.	3	(8)
12	a)	Explain how life-cycle analysis (LCA) can help engineers design more sustainable products.	3	(5)
	b)	Illustrate with one example relevant to Kerala.	3	(3)

Module 3				
13		Rapid motorization is worsening air pollution. Suggest sustainable mobility options suitable for medium-sized Kerala towns, explaining their environmental benefits.	4	(8)
14		Outline a three-step plan to move a college campus towards zero waste. Mention the role of engineers, students, and the college management in each step.	4	(8)
Module 4				
15	a)	Discuss how renewable energy technologies (solar, wind, biogas) can be scaled up in Kerala.	5	(4)
	b)	State two technical and two social challenges to their adoption.	5	(4)
16	a)	Discuss the importance of environmental policies and regulations in promoting sustainable engineering.	5	(4)
	b)	Give one national and one international example and their impact.	5	(4)



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